

An Introduction to Pre Exposure Prophylaxis (PrEP)

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# **‘PrEP’ed and Ready To Go**

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# Introduction to PrEP

Pre-Exposure Prophylaxis (PrEP) is a biomedical HIV prevention strategy that involves the daily or event-driven use of antiretroviral medication (Truvada/ (TDF/FTC)) by HIV-negative individuals at high risk of infection.



This has been a feature of treatment and care programmes since 1987 for Post Exposure Prophylaxis (PEP) and 2012 for Pre-exposure Prophylaxis (PrEP).



It has been proven effective in reducing the risk of HIV acquisition when taken consistently and correctly with an over 90% effectiveness when taken correctly and with a high level of adherence.



# Introduction to PrEP

There are two ways in which PrEP is administered: Oral daily PrEP and Event Driven (ED) PrEP

All PrEP users are encouraged to use additional prevention methods for other sexually transmitted infections (STIs) and pregnancy concurrently.

# Jamaica's Position on PrEP



- Jamaica fully supports the implementation of PrEP as part of its comprehensive HIV prevention strategy. In alignment with WHO guidelines, Jamaica emphasizes accessibility, informed choice, and targeted support for high-risk populations.
- National policies ensure that PrEP services are integrated into existing HIV prevention and care frameworks.

# Jamaica's Position on PrEP



01 Developing the Implementation guidelines Job Aides and SOPs for programme

■ 02 Expansion of PrEP services -Full integration into existing HIV prevention programmes.

■ 03 Re-sensitization for all HCWs

■ 04 Establishing of Monthly Check-in calls/meeting with relevant PrEP officers.

05 Revising the corrective actions from the PrEP audit done in 2024.

# Jamaica's Position on PrEP

- Provision of Demand Creation for PrEP (Alternatives Oral Daily PrEP verses Event Driven PrEP)
- Spreading of awareness through continued mass media campaigns and promotions of PrEP
- Reprinting and dissemination of Job aids, pamphlet and brochures
- Give support via the PrEP warm-line
- Evaluation and addition of existing sites



# WHO

# PrEP Guidelines

The World Health Organization recommends PrEP for individuals at substantial risk of HIV. Key criteria include:

- Regular HIV testing and counseling
- Monitoring of kidney function and adherence
- Integration with broader HIV prevention services
- These guidelines form the basis for Jamaica's PrEP programme implementation.

# WHO PrEP vs. Jamaica PrEP Programme

## WHO PrEP guidelines

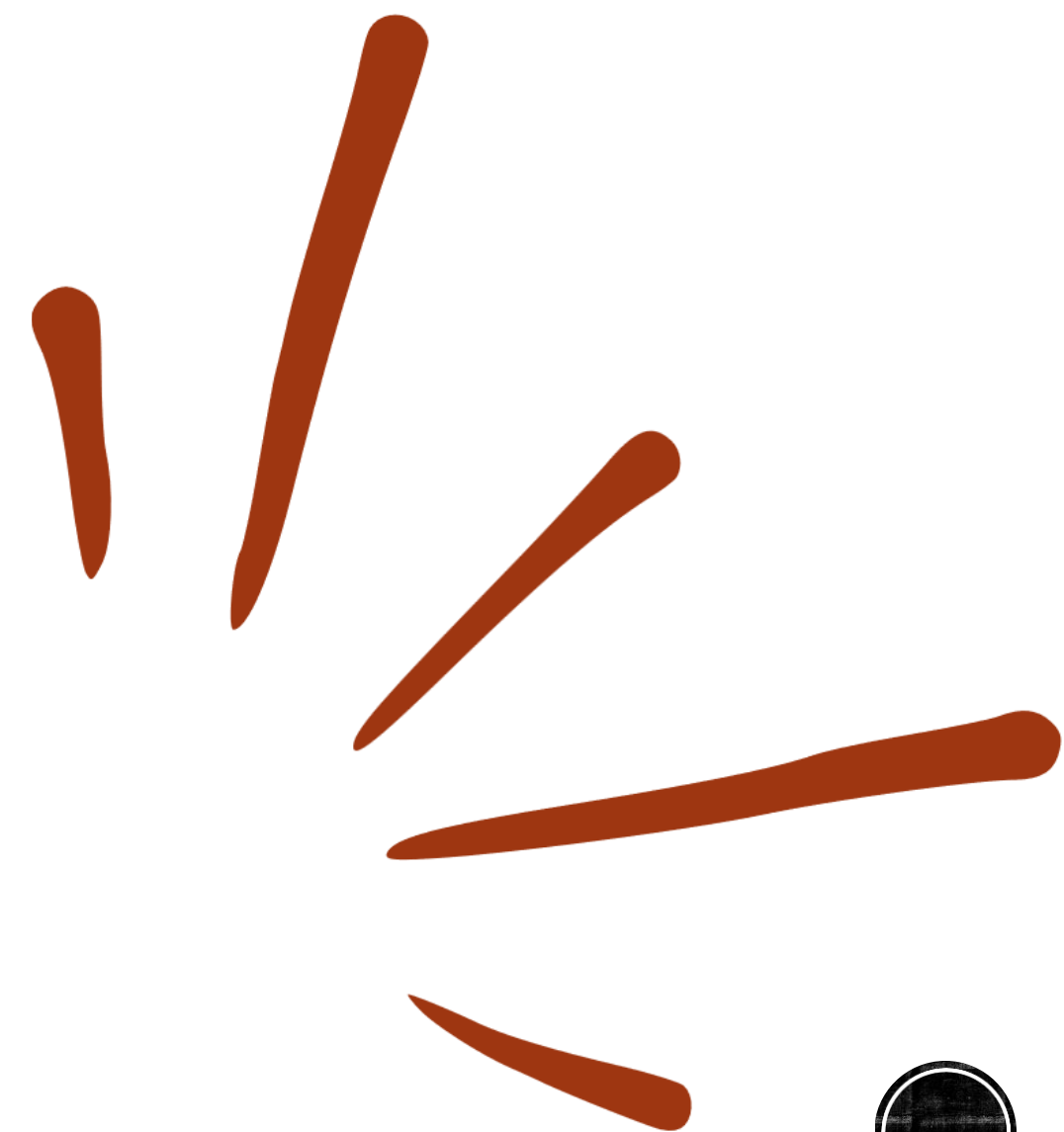
- Development of Clinical protocols and standard operating procedures to initiate a PrEP service
- Development of a PrEP curriculum and training programme for the prevention & treatment team
- Establishing a communication network for PrEP
- Clinical guidelines with the use of TDF/FTC for daily PrEP and On Demand PrEP

## Jamaica PrEP programme

- Development of Clinical protocols and standard operating procedures to initiate a PrEP service
- Development of a PrEP curriculum and training programme for the prevention & treatment team
- Establishing a communication network for PrEP
- Clinical guidelines with the use of TDF/FTC for daily PrEP and On Demand PrEP

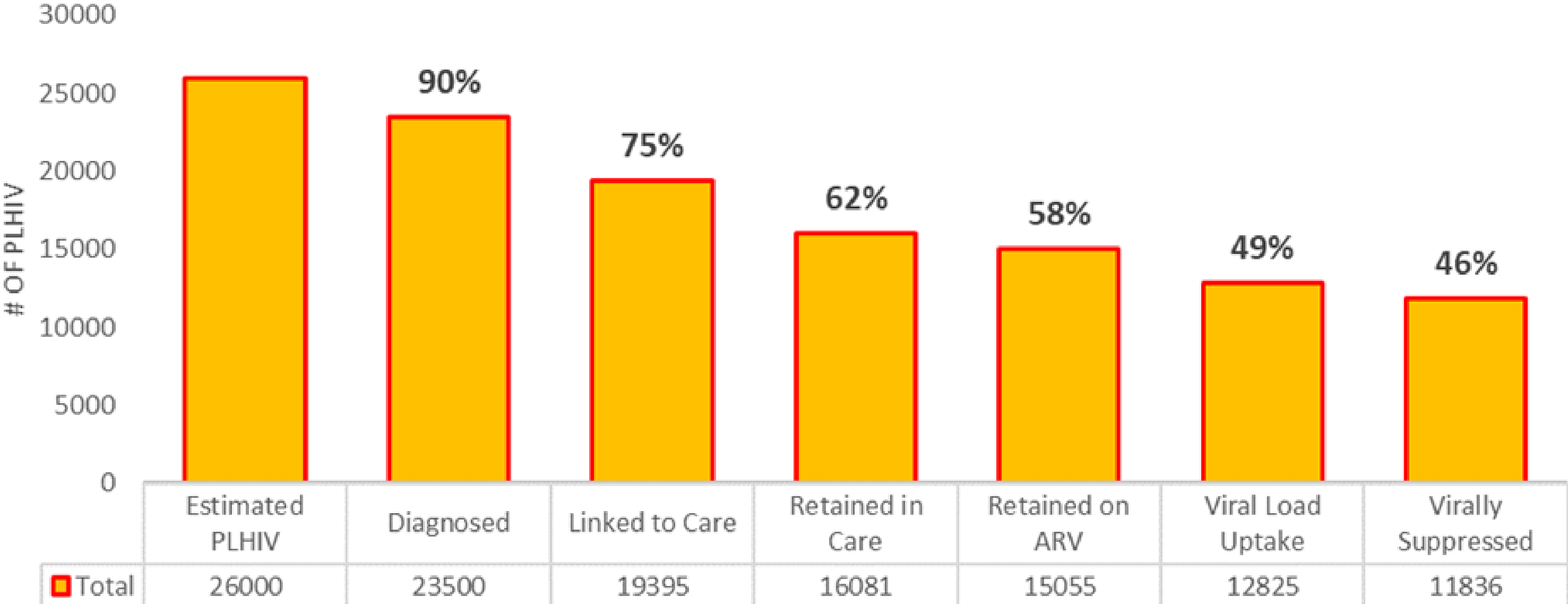
# Locations offering PrEP Services

| Parish                | Number of Facilities offering PrEP Services |
|-----------------------|---------------------------------------------|
| Clarendon             | 2                                           |
| Kingston & St. Andrew | 4                                           |
| Manchester            | 1                                           |
| St. Ann               | 2                                           |
| St. James             | 2                                           |
| St. Catherine         | 5                                           |
| St. Elizabeth         | 1                                           |
| Westmoreland          | 2                                           |
| Hanover               | 1                                           |
| Trelawny              | 1                                           |
| St. Mary              | 1                                           |
| Portland              | 1                                           |
| <b>Total</b>          | <b>23</b>                                   |



# National Treatment Cascade

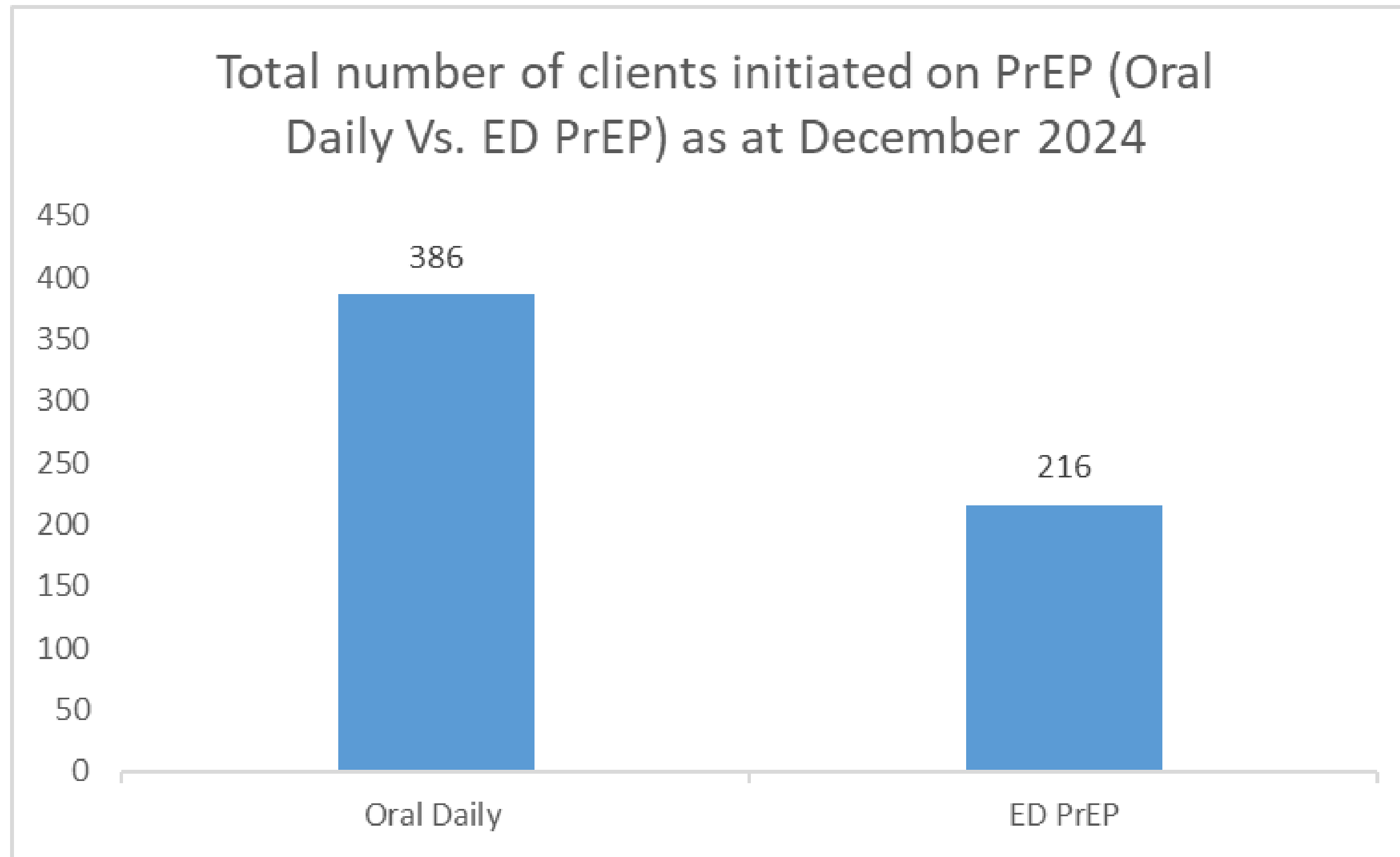
National Treatment Cascade as at December 2024



# PrEP National Targets 2024

|       |     |     |     |     |     |      |
|-------|-----|-----|-----|-----|-----|------|
| MSM   | 336 | 211 | 168 | 127 | 842 | 2215 |
| TG    | 19  | 11  | 10  | 6   | 46  | 121  |
| FSW   | 229 | 143 | 115 | 86  | 573 | 1146 |
| Youth | 80  | 50  | 40  | 30  | 200 | 399  |
| SAM   | 90  | 56  | 44  | 34  | 224 | 448  |
| SAF   | 207 | 130 | 103 | 78  | 518 | 1036 |

# Total Number of Clients Initiated On PrEP (Oral Daily vs. ED PrEP) as at December 2024



# What is Available for PrEP?

## PrEP: ARV medication as an HIV prevention method

WHO clinical guidelines recommend:



### TDF/FTC daily

as daily oral PrEP for all populations at risk of HIV acquisition<sup>1-3</sup>



### TDF/FTC on demand

as daily oral PrEP for MSM/TGW only\*<sup>1-3</sup>



### CAB LA

as IM injections given every 2 months for all populations at risk of HIV acquisition<sup>1,2</sup>



### DPV ring

for cis-gender women only as part of combination prevention approaches<sup>†4</sup>

**PrEP can significantly reduce HIV acquisition of people with significant HIV risk, including MSM, TGW and injection drug users<sup>1-3</sup>**

*Note: CAB LA PrEP is currently approved for use in the US only*

\*Currently insufficient data in cis-gender women

†As of June 2022, CAB LA is only approved for PrEP in the US

‡The DPV vaginal ring is not available in the US

ARV, antiretroviral; DPV, dapivirine; FTC, emtricitabine; IM, intramuscular; MSM, men who have sex with men; TAF, tenofovir alafenamide; TDF, tenofovir disoproxil fumarate; TGW, transgender women; US,

1. DHHS/CDC Clinical Practice Guideline: PrEP for the Prevention of HIV Infection in the United States, Updated 2021

2. Saag MS, et al. JAMA 2020;324:1651-69; 3. EACS guidelines, Version 11.0, Oct 2021

4. WHO. Consolidated guidelines on HIV prevention, testing, treatment, service delivery and monitoring: Recommendations for a public health approach. Jul

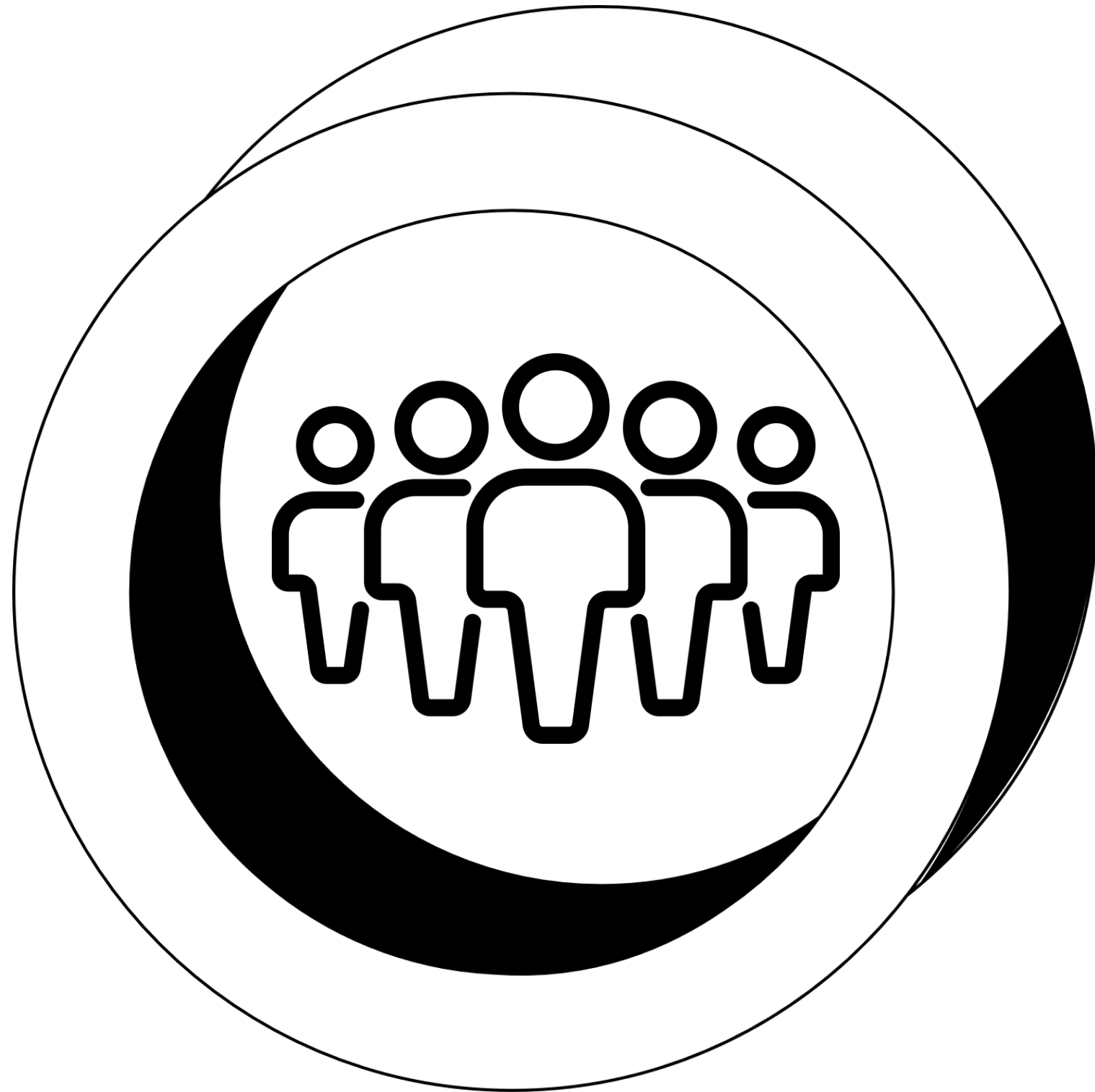
# Why is PrEP Important?

PrEP is a critical tool in the global fight against HIV:

- PrEP programmes are important interventions as they identify persons who are at risk of acquiring HIV and provide another tool to empower these individuals for reducing their personal risk.
- It offers a high level of protection when used properly.
- It complements other prevention methods such as condoms.
- It empowers high-risk populations to take control of their health.
- (PrEP) when used consistently, reduces the risk of HIV infection during sex by over 90%.



# Target Populations for PrEP



**Consider PrEP for individuals who are at substantial risk of HIV acquisition including:**

- Sexually active MSM
- Heterosexually active women and men with concurrent or high number of sex partners
- Transgender women
- Heterosexually active women and men whose partners are known to have HIV infection
- Adolescent girls, boys young women and men
- Sex workers
- Individuals using condoms inconsistently
- Anyone requesting PrEP

# Eligibility Assessment/ Screening

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**Ensure the client  
meets the following  
criteria:**

- Cisgender male or assigned male at birth
- HIV negative
- Has no signs or symptoms of Acute HIV infection
- At substantial risk of HIV through planned sexual activity

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**Obtain medical  
history**

- Sexual history (over the past 6 months)
- Other chronic illnesses

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**Provide Baseline  
testing for:**

- HIV
- Syphilis
- Hep B
- Kidney function tests (want a creatinine clearance of >60ml/min)

# Who is excluded from Initiating PrEP?

**HIV positive test results using the National HIV testing algorithm**

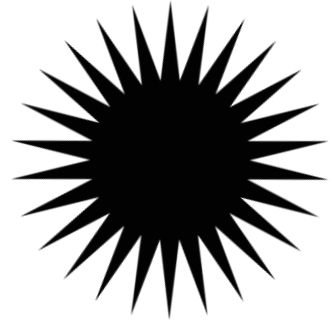
**Known exposure to HIV in the past 72 hours (Offer PEP)**

**Symptoms of Acute HIV Infection (AHI) and potential exposure within the past 14 days (defer 28 days and retest)**

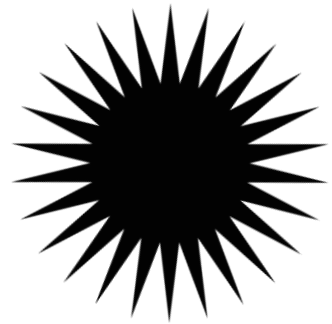
**Drug Allergy to any component of the drugs being used of oral PrEP quite uncommon**

**Creatinine clearance < 60ml/min**

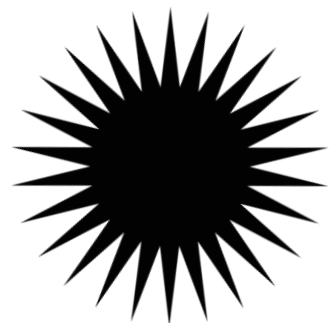
# Hepatitis B Infection



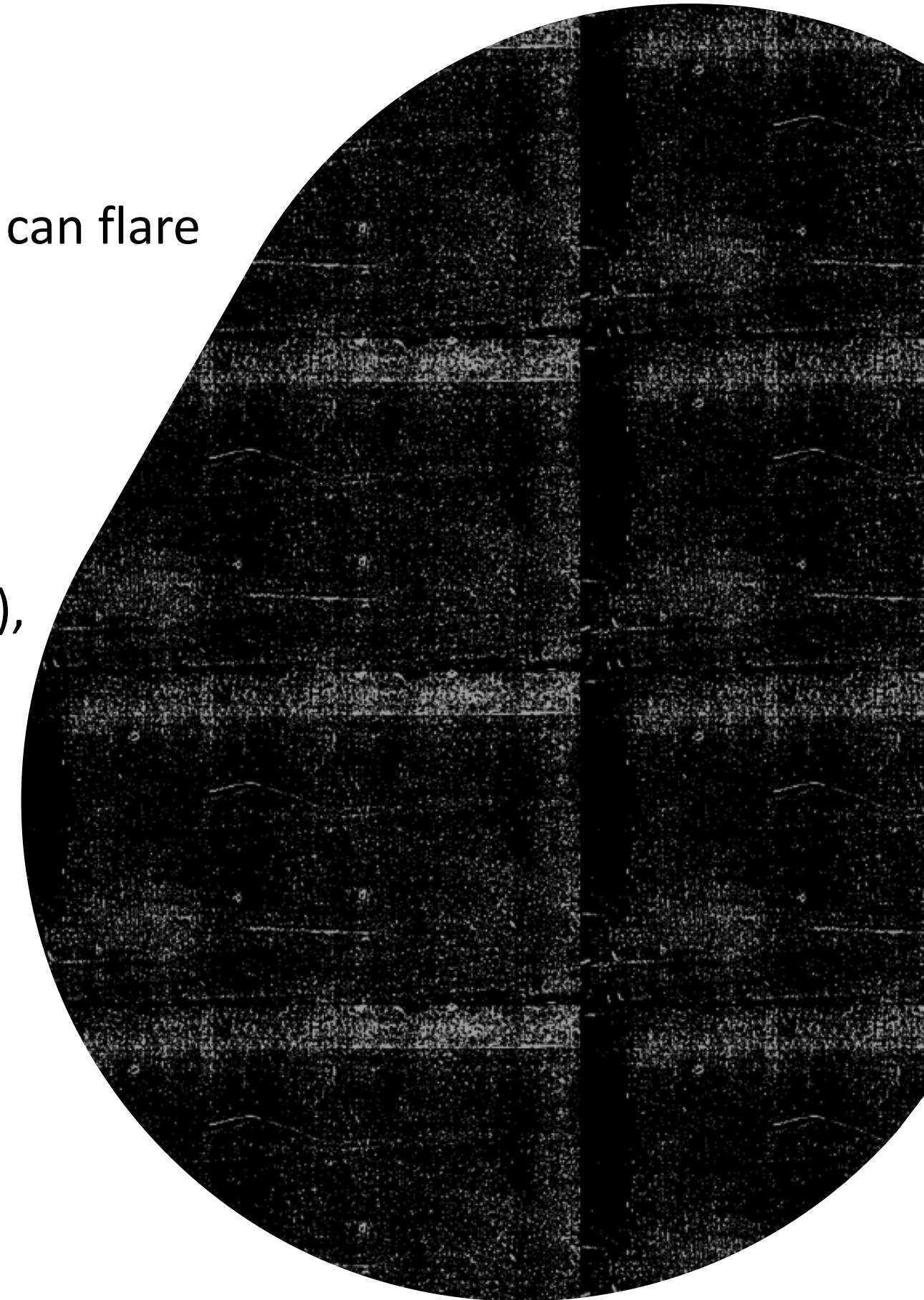
When HBV treatment is stopped, occasionally HBV infection can flare in one to three months.



Such flares are often limited to elevations in serum alanine aminotransferase (ALT) and aspartate aminotransferase (AST), although decompensation of liver function can occur.



Flares in HBV infection are treated by restarting treatment. The risk of hepatitis flares after stopping HBV treatment is higher in people with liver fibrosis.



# Hepatitis B Screening

- Hepatitis B vaccination is appropriate for people at substantial risk of HBV or HIV infection.
- Vaccination is warranted if there is no documented history of a completed vaccine series for HBV.
- PrEP can be provided whether or not HBV vaccination is available.
- Tenofovir disoproxil fumarate (TDF) is active against HBV infection at the same dose used for PrEP and is recommended by WHO

# Screening for creatinine clearance (if available)

- Estimated CrCl =  $[140 - \text{age (years)}] \times \text{weight (kg)} \times f$  where  $f=1.23$  for men and  $1.04$  for women /  $[72 \times \text{serum creatinine } (\mu\text{mol/L})]$
- PrEP is not indicated if creatinine clearance is  $< 60\text{ml/min}$ .
- ***eGFR calculators available in the app stores on smartphones***

**Creatinine Clearance Estimate by Cockcroft-Gault Equation** [Share](#)

**Input:**

Sex ☒ Male (1) ☐ Female (0.85)

Age  yr

Serum Creat  mg/dL

Weight  kg

**Result:**

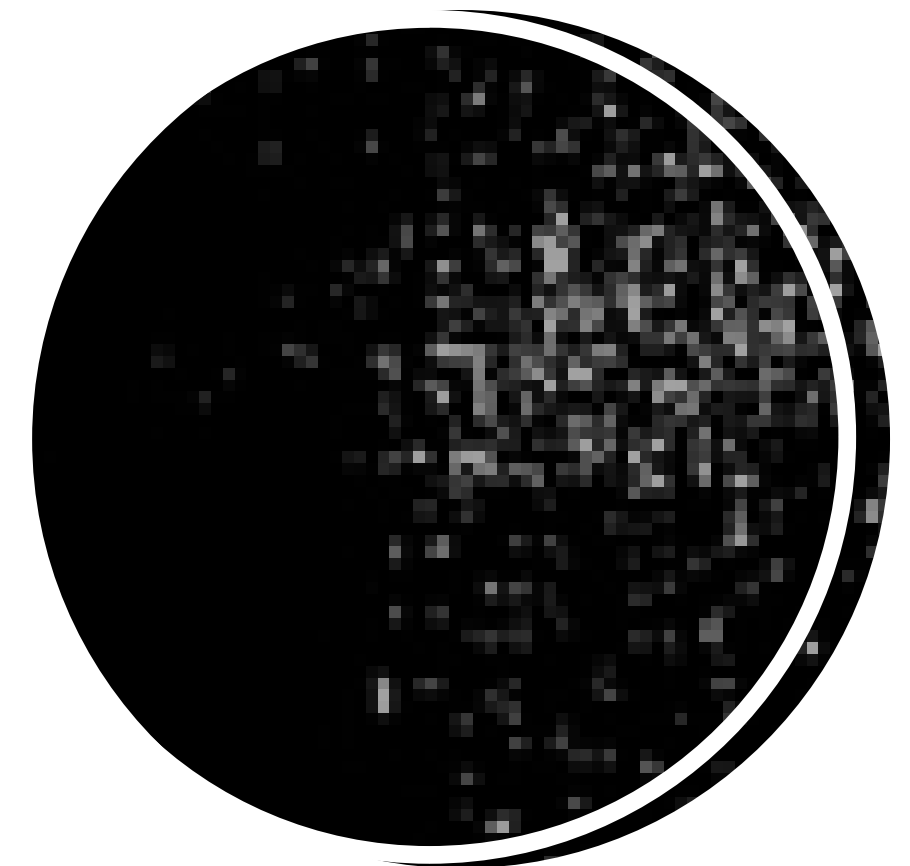
Creat Clear  mL/min

Decimal Precision:  2

[Formula](#) [Notes](#) [References](#)

$\text{CreatClear} = \text{Sex} * ((140 - \text{Age}) / (\text{SerumCreat})) * (\text{Weight} / 72)$

<http://reference.medscape.com/calculator/creatinine-clearance-cockcroft-gault>



# Managing Creatinine Elevation

one to three months, and oral PrEP can be restarted if eGFR returns to  $> 60$  ml/min.

- Stop oral PrEP if creatinine elevation is confirmed on a repeat specimen on a separate day and if estimated creatinine clearance decreases to  $<60$  ml/min.
- After oral PrEP is stopped, creatinine should be checked for another
- Consider additional causes and management of creatinine elevations if:
  - Creatinine elevation reaches more than three times the baseline
  - Renal function or creatinine elevations do not return to normal levels within three months of stopping oral PrEP
- Common causes of chronic or severe renal insufficiency include: diabetes mellitus, uncontrolled systemic hypertension, hepatitis C infection, liver failure, and pre-eclampsia during pregnancy.



# WHO Recommended Procedures for Measuring Kidney Function Of Oral PrEP Users

| Population (s)                                                                                                         | Baseline Screening                                                                           | Follow-up Screening                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Individuals 29 years and younger with no kidney-related comorbidities <sup>0</sup>                                     | Optional                                                                                     | <p>If not conducted or if baseline test is normal,<sup>1</sup> follow-up is optional until 30 years of age or if kidney-related comorbidities develop.</p> <p>If baseline test result suggests at least mild loss of kidney function,<sup>2</sup> follow-up measurements every six to 12 months are suggested.</p> |
| Individuals 30–49 years with no kidney-related comorbidities <sup>0</sup>                                              | Conduct once at initiation or during a follow-up visit one to three months after initiation. | <p>If baseline test is normal,<sup>1</sup> further screening is optional until 50 years of age or if kidney-related comorbidities develop.</p> <p>If baseline test result suggests at least mild loss of kidney function,<sup>2</sup> follow-up measurements every six to 12 months are suggested.</p>             |
| Individuals 50 years and older                                                                                         | Conduct once at initiation or during a follow-up visit one to three months after initiation. | Conduct follow-up measurements every six to 12 months.                                                                                                                                                                                                                                                             |
| Individuals of any age with kidney-related comorbidities <sup>0</sup>                                                  |                                                                                              |                                                                                                                                                                                                                                                                                                                    |
| Individuals with previous measurement of kidney function suggesting at least mild loss of kidney function <sup>2</sup> |                                                                                              |                                                                                                                                                                                                                                                                                                                    |

# First Clinic Visit/ Initiation for Oral Daily PrEP

**There are five essential components of PrEP initiation visits.**

- HIV testing and counseling (client must be confirmed negative before PrEP initiation)
- Oral PrEP screening
- Counseling and prevention messages
- Physical examination
- Prescription

# Same Day Oral PrEP Initiation

Same day Initiation is the gold standard for Oral PrEP programs. The screening and oral PrEP initiation should be combined to improve access.

For most clients, PrEP can be initiated the same day. However, in some scenarios, as outlined below, deferred PrEP initiation is recommended.

Clients must meet four criteria to begin PrEP use. They must be:

- **HIV negative**
- **Not indicated for PEP or suspected of having AHI**
- **Requesting PrEP or indicated for PrEP use**
- **Free from contraindications for use of their chosen oral PrEP regimen**

# Prescribing Oral PrEP at Initiation

At initiation, client could be prescribed:

- a single bottle and scheduled for a one-month follow-up visit, then
- multiple bottles and scheduled for a three-month follow-up visit
- Stress the importance of condoms for 7 days after initiation
- A one-month follow-up visit is ideal to provide counseling on effective use, address any side effects, and rule out AHI at initiation and review blood investigations (serum Creat, Hep B, C).

Schedule the client's next visit a couple days to a week before the pill supply will run out based on daily use



# Follow up visit for Oral Daily PrEP



At follow-up visits, clients can receive multiple bottles of PrEP.



If the client prefers fewer bottles because they lack a place to discretely or safely store it, fewer bottles can be prescribed/dispensed

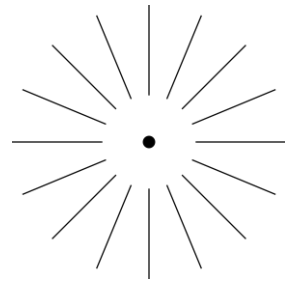


Schedule the client's next visit a week before the pill supply will run out based on daily use, at least every three months.

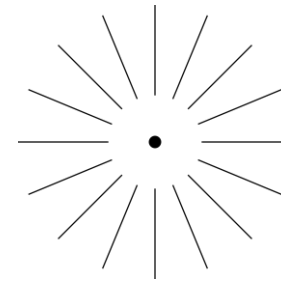


HIV and STI testing done at each follow up visit.

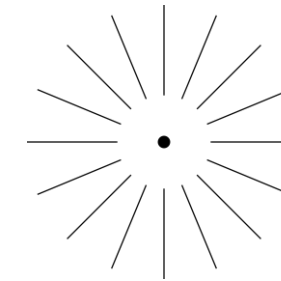
# Restarting Oral Daily PrEP



Some follow-up visits  
may be restart visits



Clients can and should  
use oral PrEP when  
they need it



Restart visits have the  
same components of  
regular follow-up visits.

# Oral Daily PrEP

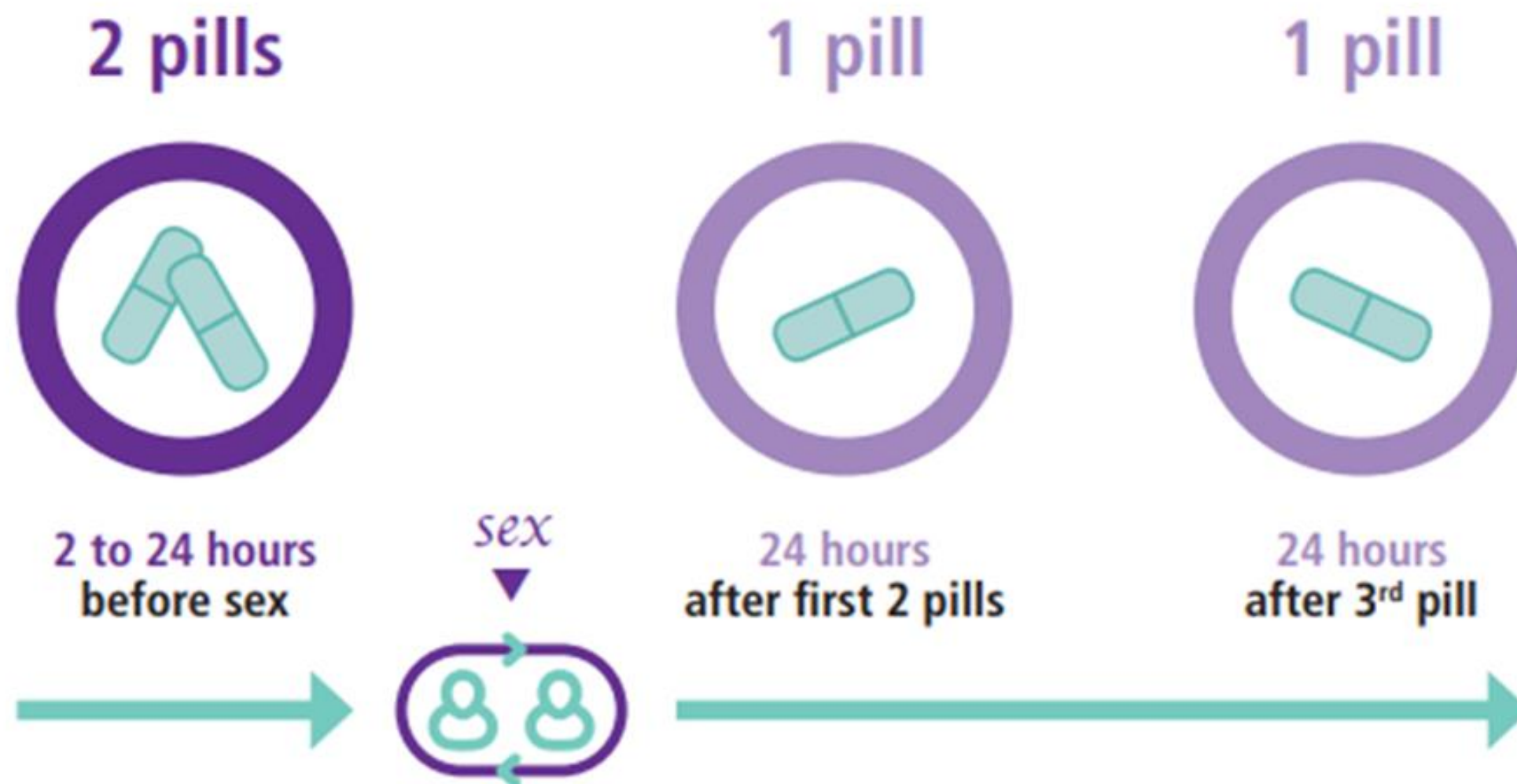


Oral Daily PrEP involves taking one pill every day. This regimen maintains steady levels of antiretroviral drugs in the bloodstream, providing continuous protection against HIV.



It is recommended for individuals with frequent exposure to HIV, such as those in serodiscordant relationships or with multiple partners.

# Event-Driven PrEP



- Event-Driven, or 'On-demand' PrEP, is an alternative regimen for individuals with less frequent HIV exposure.
- This method involves taking a double dose, 2-24 hours before anticipated exposure, followed by a single dose 24 and 48 hours after the event. It requires careful planning and adherence to the dosing schedule.

# Screening Process for Event Driven PrEP

Before initiating ED PrEP, candidates undergo a somewhat similar screening process however this is much more concise:

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Explain the purpose and effectiveness of ED-PrEP along with possible side effects

Confirming HIV-negative status (documented by recent HIV testing done in the past 7 days.

No signs or symptoms of AHI

Assessing kidney function and overall health. (this test is only required in eligible cisgender males over 30 yrs or has a chronic illness affecting the kidney)

Screening for syphilis and evaluating overall risk factors.

# Safety and Side Effects

- PrEP is considered safe for everyone, with few side effects.
- Some users may experience mild symptoms such as nausea, headache, or slight gastrointestinal discomfort during the initial weeks of use.
- Ongoing monitoring ensures any adverse effects are promptly managed.



# Challenges

While PrEP is a powerful tool, challenges remain:

- Ensuring high adherence levels
- Addressing stigma associated with its use
- Expanding access to at-risk populations



# Challenges

## System Challenges

Turn around time for Results especially (U& E)

Lengthy wait time

Staffing challenges

## Patient Challenges

Persons slowly gravitating towards the idea of PrEP

Education level of patients

Fear/ stigma of PrEP

Cultural Behaviors (not a pill taking country)

# Future Plans/Way Forward

**Finalizing of  
Implementation  
guidelines Job Aides and  
SOPs for PrEP programme**

**Continue Re-  
sensitization and  
education for the  
general staff and all  
HCWs**

**PrEP survey to asses  
PrEP usage of clients**

**Continue Monthly Check-  
in Meetings with the  
relevant staff members**

**Continue to provide  
alternatives: Oral Daily  
PrEP verses Event Driven  
PrEP**

# Future Plans/Way Forward

- Explore the possibility of PrEP injectable Pilot
- Continue Spreading awareness through media and promotion of PrEP
- Continued reprinting and dissemination of Job aids, pamphlets and brochures
- Continued support via the PrEP warm-line
- Expansion of PrEP services to new sites
- Evaluation of existing sites.

# Summary

- PrEP is an effective HIV prevention strategy when used correctly.
- Both Oral Daily and Event-Driven regimens offer tailored protection based on risk patterns.
- The use of additional prevention methods for other sexually transmitted infections (STIs) is to be emphasized and encouraged.
- Jamaica's PrEP programme aligns with WHO guidelines to ensure safe and effective implementation.
- While there are still challenges, ongoing efforts are required to improve adherence, reduce stigma, and expand access.



# E- Learning Resource

The Oral PrEP eLearning course created through a collaboration between WHO and Jhpiego is a self-paced, free eLearning course available at [hivoralprep.org](http://hivoralprep.org).

The training prepares clinicians to provide both daily oral PrEP and ED-PrEP.

## Oral PrEP eLearning Resource Package:

A Collaboration Between WHO and Jhpiego



The Oral PrEP eLearning Resource Package (eLRP) introduces health care providers to Oral Pre-Exposure Prophylaxis (PrEP) for HIV infection. The eLRP includes eLearning courses for clinicians with an accompanying Oral PrEP Implementation



# References

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- WHO Implementation tool for pre-exposure prophylaxis (PrEP) of HIV infection
- Guideline on when to start antiretroviral therapy and on pre-exposure prophylaxis for HIV. Geneva: World Health Organization; 2015
- <https://iris.who.int/bitstream/handle/10665/255890/WHO-HIV-2017.19-eng.pdf>
- <https://www.prepwatch.org/> Retrieved on August 19, 2024



Thank You!

**Questions & Discussion**